

Course Code : ECT 359-4

KQLR/RS – 24 / 1179

**Sixth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

INTRODUCTION TO IOT

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Due credit will be given to neatness.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) For a Smart City Transportation system Use Case, recommend an architecture with details and justification. 7(CO1,5)
 - (b) As an IoT Application Developer, select an Access Technology for applications. Justify. 4(CO1,5)
2.
 - (a) Explain in detail TCP / IP protocol Stack. 4(CO1)
 - (b) Describe the steps to execute a bash script after loading the desktop environment on RPi. 3(CO2)
3.
 - (a) Write a bash script to find the smallest and largest number from the array (24 27 84 11 99). 5(CO2)
 - (b) Write a bash script to add executable permission to a python script “demo.py” and execute the python script with two arguments from an arrays having length 8 and 6. 5(CO1,3,4)
4.
 - (a) Write a python script to perform moving average of five elements of an array X = [1 2 3 4 5 6 7 8 9]. 5(CO1,3,4)
 - (b) Write a python script to read an analog water level sensor deployed in a remote area using pi to control a pump locally driven by a relay, PUMP should be ON below 75% and otherwise OFF. 7(CO1,3,4)

KQLR/RS-24 / 1179

Contd.

5. (a) Write a python script to interface Pi and Accelerometer ADXL 345 to measure speed and acceleration using I2C. 5(CO1,3,4)
- (b) Compare UART and SPI protocol with examples. 4(CO4)
6. (a) Identify the challenges of implementing an IoT Use case for Remote Patient Diagnosis system for chronic diseases in India. How would you overcome those ? Justify. 7(CO1,5)
- (b) Explain the COAP protocol with an example. 4(CO1)

